Configurable Hemisphere Environment for Spatialised Sound

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Abstract

This paper reports on a cross-disciplinary research initiative in spatialised audio at the University of Wollongong. It describes a multi-channel sound environment in the School of Music and Drama and its background. Some of the creative tools developed by software engineers working from the Whisper Research Laboratory in the Faculty of Informatics are presented together with strategies for usage by sound designers, composers and installation artists working with spatialised sound.

1 CHESS – a Configurable Hemisphere Environment for Sound Spatialisation

The CHESS system is an immersive sound projection system built to audition new forms of sound localisation. It consists of a loudspeaker array mounted in a spherical structure that can be configured to audition sound in a variety of digital formats. The user is able to configure many speaker positions including standard configurations and other customised speaker placements.

The CHESS system was built to serve digital audio research undertaken by engineering specialists from the Faculty of Informatics. It is used by Informatics staff and a PhD candidate from the Faculty of Informatics.

The system is located in the sound studios of the Faculty of Creative Arts where it is accessed by artists working in sound allowing them to monitor work produced in 3D audio. Within the Faculty of Creative Arts users currently include staff from the School of Music and Drama and the School of Journalism and Creative Writing. In time this will also include the School of Art and Design as new Post-Graduate programs are introduced.

The ability to reconfigure speakers in a variety of positions enables artists to create new simulated environments quickly and and experiment with them in production. Speakers are attached to hinged brackets mounted on the frame. These brackets can be positioned in a variety of ways e.g. raised, lowered or swivelled horizontally, allowing speakers to be repositioned around, above and below the listener.

2 History of CHESS

The CHESS system was set up with the help of an ARC RIBG grant in 2001. This was an initiative of the Digital Media Centre which sought to support new developments in interactive television. The initiative involves a collaborative research effort between the 3D-audio team from the Informatics Faculty led by Ian Burnett and staff in the sound, composition, music and production program in the Faculty of Creative Arts led by Stephen Ingham. In 2003 a further RIBG grant secured additional digital recording equipment and signal processing hardware developed by Lake Technology. Lake also made substantial financial contributions.

Didier Balez, a sculptor who works with steel and ceramics, and technician in the Visual Arts workshop in the Faculty of Creative Arts, constructed a customised steel structure to house the CHESS system working under the supervision of Ian Burnett and Stephen Ingham. The system was officially launched by the Pro-vice Chancellor (Research) in August 2003 at the opening of Sonic Connections, a 2-day showcase of research and creative work at the Faculty of Creative Arts [1].

The CHESS system uses a Pentium 4 PC communicating with a Mac G4 via a LAN. The PC provides a user interface for spatilising audio produced by the G4.

Spatilisation takes place using a java program that runs on the PC which transmits a stream of spatial coordinates to the G4. The G4 processes the audio which is fed to amplifiers and loud speakers. Figure 2 describes the current configuration of CHESS.

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Figure 1. Configurable Hemisphere Environment for Sound Spatialisation (CHESS)

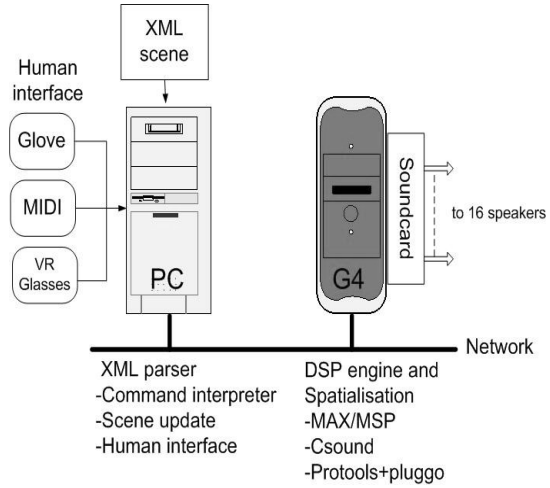


Figure 2. Current processor configuration used in CHESS

The java application running on the PC codifies these coordinates as a stream of spatial data called a 'scene'. Scenes are described in XML format and communicated to the G4 using the UDP protocol. The role of the G4 is to run signal processing algorithms to place sounds in a virtual auditory space.

These algorithms include:
Spatialisation: Ambisonics (1st to 4th order) or VBAP;

Delays and attenuations to account for the distance of sound sources and the properties of the medium;

Decorrelation, to create broad sound that can be used for beach or waterfall sound sources;

Calculation of reflections and obstructions to compute early reverberation;

Simulation of late reverberation by a Feedback Delay Networks algorithm controlled by perceptual parameters.

Using a PC to define the 3D scene and transmitting this to the G4, frees G4 resources for audio signal processing in user applications. Currently 24 sound sources can be spatialised by the 867MHz G4.

Figure 2 shows a sound card (Digi-001) connected to 16 speakers (1029a Genelec) which make up the hardware associated with the CHESS system. A Soundfield microphone is used to capture real 3D sound environments that can then be played back accurately in CHESS or be mixed with other sounds and environments.

3 Key Developments - Informatics

Research done by staff in Informatics over the past three years involves extensive publication and commercially sensitive research. Areas of published research include:

investigation of the apparent source width and shape of 3D sound [2][3][5][6];

Development of an XML scheme for describing 3D audio scenes [4][8];

development of a Java3D application by Mark O'Dwyer for parsing, rendering and visualising 3D audio XML scenes [2];

development of a java application for describing 3D audio scenes using XML [7][8];

development of Max objects for 4th order Ambisonics spatialisation and 5.1 panning (Potard, Informatics);

development of Max external objects to produce wide sound sources (Potard, Informatics);

development of software enabling B-format recordings to be played back in 5.1 and other spatialisation formats up to 16-Channel;

development of a set of pluggins to perform up to 4th order Ambisonics spatialisation on up to 16 speakers from within Protools LE;

developed tools for virtual positioning of sound objects using data glove and virtual reality glasses (O'Dwyer, Informatics);

comparisons made between the precision and source localisation characteristics of different spatialisation algorithms;

Figure 3 shows the Max panning object for the CHESS system.

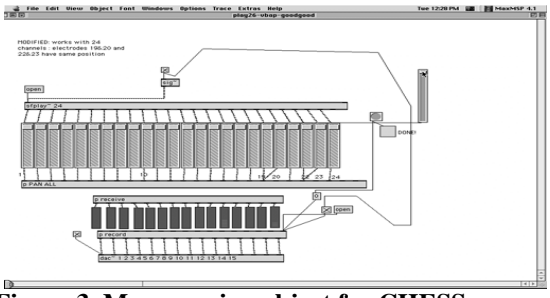


Figure 3. Max panning object for CHESS

3 User software

Currently Max/MSP is the principal tool for signal processing on the G4. Other musical applications have also been used including Csound and Protools using the Pluggo plug-in architecture. The chosen spatialisation algorithm is Ambisonics as it provides a format which is independent of a particular speaker configuration. Thus created scenes can be played on a different speaker configuration with out the need to regenerate scenes. The sweet spot and stability of sound sources are greatly improved using higher order ambisonics (HOA). Using 4th order ambisonics 25 signals are used to represent the sound field compared to 4 for 1st order (i.e. B-format).

Alternatively, the Max object, called VBAP (an acronym for Vector Based Amplitude Panning) can also be used to perform spatialisation of sound sources. Spatialisation using VBAP is less precise than ambisonics.

The Protools Plugin, called Pluggo RTAS (an acronym for Real Time Audio Suite) is relevant for Max users. It allows the CHESS system to play audio encoded in ambisonic B format by converting it to formats playable in any of the speaker configurations available in CHESS.

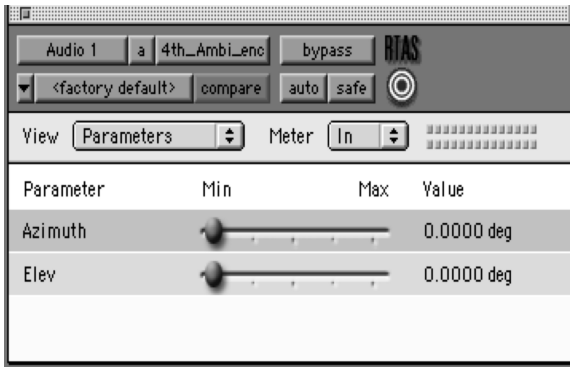


Figure 4. Protools Pluggo Plugin

This application has been used by Gareth Jenkins, a Master's candidate working under the supervision of John Scott in the School of Creative Writing, in collaboration with Guillaume Potard, to create a narrative work lasting 45 min entitled "*Owen's Waiting"*. It was produced in Protools and encoded in 16 channels on the CHESS system using Pluggo plug-in. It was presented at Newcastle ElectroFringe in October 2003.

The first postgraduate users of the CHESS system from the Sound Composition Music and Production program are Masters candidates Damien Lock and Aaron Hull. Lock will investigate possible use of spatialisation in animation post production packages like 3D Studio max or Maya "Physics Engine" Plugins while Hull will further investigate possibilities for spatialisation in live performance using the Protools Pluggo Plugin.

Csound has also been used to produce spatialised audio with CHESS. Csound can also be used as a Csound object in Max/MSP and as a means for creating VST pluggins. This allows users familiar with Cubase VST to work with the CHESS system.

Csound was used by Greg Schiemer and Guillaume Potard for sonification of brain data in a work entitled "*Neural Dialogues*" which was submitted for the ICAD project entitled "Listening to the Mind Listening" [9]. Due to bandwidth limitations of the G4, it was necessary to work in two stages, one using Csound to create a 24-channel sound file which was panned using a Max program consisting of 24 VBAP object.

The Lake Huron Digital Audio Convolution workstation acquired in 2003 will overcome these bandwidth limitations. The Huron offers several spatialisation tools including binscape used to monitoring 3D sound environments in binaural headphones using Lake Theatrephone. The current Huron configuration will support 16 channels and with additional DSP cards will support up to 256 in realtime.

Currently CHESS is used as part of the "Listening to the Mind Listening" project as the main processing station for concert submissions [10]. The Lake Huron and custom Max/MSP patches are used to process submissions to a binaural format and for a custom 15-speaker array in the Studio, Opera House. Using the flexibility of CHESS, a scaled down replica of this speaker configuration has been set-up.

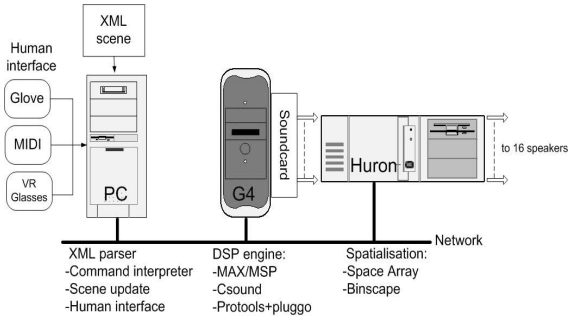


Figure 5. Projected future configuration using the PC/G4 with Lake Huron

5 Projections for the future

Although CHESS is designed as a reconfigurable speaker array it provides a relatively small audition space that is ideally suited for a single listener. The permanent nature of the frame supporting the speakers makes its relocation impractical. In its current state it does not address the need to project spatialised audio in public performance venues.

Further funding has been sought to extend the CHESS system for use in concerts and sound installations. The current system will continue to be used as a research and production space. Its extension will support presentation using spatialised audio in public exhibition spaces that provide a suitable location for larger audiences. This will also accommodate developments in new media performance where sound is an interactive partner with visual, haptic and kinaesthetic media.

The CHESS system represents a collaboration that crosses disciplinary boundaries. New audio tools have been developed and will continue to support digital audio research that leads the development of new standards. Development of new software extends the capabilities of sound production tools used by various composers and sound designers irrespective of their technical expertise or musical aesthetic. CHESS has encouraged new research along with unexpected and mutually beneficial outcomes.

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